LEVEL - VI

SINGLE ANSWER QUESTIONS

1. Let $f(x)=\underset{n\to\infty}{Lim}\dfrac{\left(x^2+2x+3+\sin\pi x\right)^n-1}{\left(x^2+2x+3+\sin\pi x\right)^n+1}$, then

A) $f(x)$ is continuous and differentiable for all $x\in R$

B) $f(x)$ is continuous but not differentiable for all $x\in R$

C) $f(x)$ is discontinuous at infinite number of points.

D) $f(x)$ is discontinuous at finite number of points.

2. Let

$$f(x)=\underset{h\to 0}{Lim}\frac{\left(\sin(x+h)\right)^{\ln(x+h)}-\left(\sin x\right)^{\ln x}}{h}$$

then $f\left(\dfrac{\pi}{2}\right)$ is

A) equal to 0 B) equal to 1

C) $\ln\dfrac{\pi}{2}$ D) non existent

3. Let $g(x)=\begin{cases}\dfrac{x^2+x\tan x-x\tan 2x}{ax+\tan x-\tan 3x}; & x\neq 0\\ 0 & ; \ x=0\end{cases}$.

If $g'(0)$ exists and is equal to non zero value b, then $\dfrac{b}{a}$ is equal to

A) $\dfrac{7}{13}$ B) $\dfrac{7}{26}$ C) $\dfrac{7}{52}$ D) $\dfrac{5}{52}$

4. If $\underset{n\to\infty}{Lim}\dfrac{n.3^n}{n(x-2)^n+n.3^{n+1}-3^n}=\dfrac{1}{3}$ where $n\in N$, then the number of interger(s) in the range 'x' is

A) 3 B) 4 C) 5 D) infinite

5. Let

$$f(x)=\frac{e^{\tan x}-e^x+\ln(\sec x+\tan x)-x}{\tan x-x}$$

be a continuous function at x = 0. The value of $f(0)$ equals

A) $\dfrac{1}{2}$ B) $\dfrac{2}{3}$ C) $\dfrac{3}{2}$ D) 2

6. $$f(x)=\begin{cases}\dfrac{\cos^{-1}\left(1-\{x\}^2\right)\sin^{-1}\left(1-\{x\}\right)}{\sqrt{2}\left(\{x\}-\{x\}^3\right)}, & for\ x\neq 0\\ \dfrac{\pi}{2} & , for\ x=0\end{cases}$$

where $\{.\}$ denotes fractional part of x, then, at $x=0,\ f(x)$, is

A) continuous B) only left continuous

C) only right continuous

D) neither left continuous nor right continuous

7. The value of

$$\underset{x\to\frac{\pi}{2}}{\text{Lim}}\left(2^{\frac{1}{\cos^2 x}}+3^{\frac{1}{\cos^2 x}}+4^{\frac{1}{\cos^2 x}}+5^{\frac{1}{\cos^2 x}}+6^{\frac{1}{\cos^2 x}}\right)^{2\cos^2 x}$$

is

A) 1 B) 6 C) 36 D) $\dfrac{1}{36}$

8. If f(x) is continuous function $\forall x\in R$ and the range of $f(x)$ is $\left(2,\sqrt{26}\right)$ and

$g(x)=\left[\dfrac{f(x)}{c}\right]$ is continuous $\forall x\in R$,

then the least positive integral value of c is (where [.] denotes the greatest integer function)

A) 2 B) 3 C) 5 D) 6

9. The function $f(x)$ defined by

$$f(x)=\begin{cases}\log_{(4x-3)}\left(x^2-2x+5\right), & \frac{3}{4}<x<1\ and\ x>1\\ 4, & x=1\end{cases}$$

A) is continuous at $x = 1$

B) is discontinuous at $x = 1$ since $f\left(1^{+}\right)$ does not exist though $f\left(1^{-}\right)$ exists.

C) is discontinuous at $x = 1$ since $f\left(1^{-}\right)$ does not exist though $f\left(1^{+}\right)$ exists.

D) is discontinuous at $x = 1$ since neither $f\left(1^{+}\right)$ nor $f\left(1^{-}\right)$ exists.

10. $f(x) = \max\left\{\frac{x}{n}, |\sin \pi x|, n \in N\right\}$ **has maximum points of non-differentiabilityu for** $x \in (0,4)$**, then**

A) maximum value of n in more than 4.5
B) least value of n is more that 3.5
C) maximum value of n is less than 4.5
D) least value of n is less than 3.5

11. $\lim\limits_{x\to\infty} \frac{\log_{10}(x+2)+[x]+2}{\left[x+\sum\limits_{r=0}^{\infty}\left(\frac{1}{2}\right)^{r}\right]}$ **(where [.] is G.I.F)=**

A) 1 **B) 0**
C) 2 **D) does not exist**

12. **If xy > 0 and** $(x+y)\frac{y}{2} = \frac{-3}{2}$, $x+y+\frac{y}{x} = \frac{1}{2}$ **then** $\lim\limits_{n\to\infty}\sum\limits_{r=0}^{n}\left(\frac{x}{y}\right)^{n} =$

A) 1 **B) 3** **C) 0** **D)** $\frac{3}{2}$

13. **Let** $f : R \to R$ **be a continuous into function satisfying** $f(x) + f(-x) = 0, \forall\, x \in R$. **If** $f(-3) = 2$ **and** $f(5) = 4$ **in** $[-5,5]$**, then the equation** $f(x) = 0$ **has**

A) exactly three real roots
B) exactly two real roots
C) atleast five real roots
D) atleast three real roots

14. **Let** x_n **be defined as** $\left(1+\frac{1}{n}\right)^{n+x_n} = e$**, then** $\lim\limits_{n\to\infty} x_n$ **equals**

A) 1 B) $\frac{1}{2}$ C) $\frac{1}{e}$ D) 0

15. **Let** $f(x)$ **be continuous for all** $x \in R$ **except at x = 0 and**
$f'(x) < 0\ \forall\ x \in (-\infty, 0)$;
$f'(x) > 0\ \forall\ x \in (0, \infty)$. **Let**
$\lim\limits_{x\to 0^{+}} f(x) = 3,\ \lim\limits_{x\to 0^{-}} f(x) = 4,\ f(0) = 5$.
Then the image of $(0,1)$ **about the line** $y \lim\limits_{x\to 0} f\left(\cos^{3} x - \cos^{2} x\right) = x \lim\limits_{x\to 0} f\left(\sin^{2} x - \sin^{3} x\right)$ **is**

A) $\left(\frac{12}{25}, \frac{-9}{25}\right)$ B) $\left(\frac{11}{25}, \frac{-9}{25}\right)$

C) $\left(\frac{16}{25}, \frac{-8}{25}\right)$ D) $\left(\frac{24}{25}, \frac{-7}{25}\right)$

MULTI ANSWER QUESTIONS

16. **Which of the following statements is/are true?**

A) If 'f' is differentiable at x = c then $\lim\limits_{h\to 0}\frac{f(c+h)-f(c-h)}{2h}$ exists and equals $f'(c)$

B) Given a function 'f' and a point 'c' in the domain of f, if $\lim\limits_{h\to 0}\frac{f(c+h)-f(c-h)}{h}$ exists then the function is differentiable at $x = c$

C) Let $g(x) = \begin{cases} x^{2}\sin\frac{1}{x^{2}}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ then g'exists

D) Let $g(x) = \begin{cases} x\sin\frac{1}{x^{2}}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ then g'exists and is continuous.

17. **Which of the following function(s) not defined at** $x = 0$ **has/have removable discontinuity at** $x = 0$**?**

A) $f(x)=\dfrac{1}{1+2^{\cot x}}$

B) $f(x)=\cos\left(\dfrac{|\sin x|}{x}\right)$

C) $f(x)=x\sin\left(\dfrac{\pi}{x}\right)$ D) $f(x)=\dfrac{1}{\ln|x|}$

18. **f(x) = min {1, cos x, 1-sin x}, $-\pi \le x \le \pi$, then**
A) f(x) is not differentiable at 0
B) f(x) is differentiable at $\pi/2$
C) f(x) has local maxima at 0
D) f(x) local maximum at x= $\pi/2$

19. **If f(x) = min {tan x, cot x}, then**

A) f(x) is discontinuous at $x=0, \dfrac{\pi}{4}, \dfrac{5\pi}{4}$

B) f(x) is continuous at $x=0, \dfrac{\pi}{2}, \dfrac{3\pi}{2}$

C) Range of $f(x)$ is $(-\infty,-1]\cup[0,1]$

D) f(x) is periodic with period π

20. **The function f(x) = $\left||e^x-1|-1\right|$ is**
A) continuous for all x
B) differentiable for all x
C) not continuous at x = 0, ln 2
D) not differentiable at $x = \ln 2$

21. **Given a real valued function *f* such that**

$$f(x)=\begin{cases}\dfrac{\tan^2\{x\}}{\left(x^2-[x]^2\right)} & for\ x>0\\ 1 & for\ x=0\\ \sqrt{\{x\}\cot\{x\}} & for\ x<0\end{cases}$$ **, where [x] is**

the integral parrt and {x} is the fractional part of x, then

A) $\lim\limits_{x\to0^+} f(x)=1$

B) $\lim\limits_{x\to0^-} f(x)=\cot 1$

C) $\cot^{-1}\left(\lim\limits_{x\to0^-} f(x)\right)^2=1$

D) $\tan^{-1}\left(\lim\limits_{x\to0^+} f(x)\right)=\dfrac{\pi}{4}$

22. **If $f(x)=\text{sgn}\left(x^2-ax+1\right)$ has maximum number of points of discontinuity, then**
A) $a\in(2,\infty)$ B) $a\in(-\infty,2)$
C) $a\in(-2,2)$ D) $a\in[-2,2]$

23. **If** $f(x)=\begin{cases}x^2\left(\text{sgn}[x]+\{x\}\right), & 0\le x<2\\ \sin x+|x-3|, & 2\le x<4\end{cases}$,

where [] and {} represent the greatest integer and the fractional part function, respectivley.

A) $f(x)$ is differentiable at x = 1

B) $f(x)$ is continuous but non-differentiable at x = 1

C) $f(x)$ is non-differentiable at x = 2

D) $f(x)$ is discontinuous at x = 2.

24. **Given** $f(x)=\sum\limits_{r=1}^{n}\left(x^r+x^{-r}\right)^2$ **;** $x\ne\pm1$ **and**

$$g(x)=\begin{cases}\lim\limits_{n\to\infty}\left(\left(f(x)-2n\right)x^{-2n-2}\left(1-x^2\right)\right) & for\ x\ne\pm1\\ -1, for\ x=\pm1\end{cases}$$

then $g(x)$
A) is discontinuous at x = –1
B) is continuous at x = 2
C) has a removable discontinuity at x = 1
D) has an irremovable discontinuity at x = 1

COMPREHENSION QUESTIONS

PASSAGE : 1

$$f(x)=\begin{cases}x+a & x<0\\ |x-1| & x\ge0\end{cases}$$

$$g(x)=\begin{cases}x+1 & x<0\\ (x-1)^2+b & x\ge0\end{cases}$$

Where a and b are non – negative real numbers

25. **The value of a, if $(gof)(x)$ is continuous for all real x , is**

A) – 1 B) 0 C) 1 D) 2

26. The value of b , if $(gof)(x)$ is continuous for all real x , is

A) – 1 B) 0 C) 1 D) 2

27. For these values of a and b , $(gof)(x)$ for all x $\in(-1,1)$ *is*

A) Even B) odd

C) neither even nor odd

D) symmetrical about x – axis

PASSAGE : 2

Two functions $f(x)\,\&\,g(x)$ are defined as,

$$f(x)=\begin{cases}[h(x)]-\left\{\dfrac{h(x)}{2}\right\}, & \text{for } x\in \text{domain of h}\\ 0 & \text{, for } x\notin \text{domain of h}\end{cases}$$

$$g(x)=\begin{cases} sgn\ h(x), & \text{for } x\in \text{domain of h}\\ 0 & \text{, for } x\notin \text{domain of h}\end{cases}$$

Where,

$$h(x)=\frac{1}{\sqrt{b-a}}\cdot\frac{\sqrt{\frac{b-a}{a}}\sin 2x}{\sqrt{1+\left(\sqrt{\frac{b-a}{a}}\sin x\right)^2}}\cdot\sqrt{a+b\tan^2 x}$$

for $b>a>0$ & $[.]$ denotes G.I.F & $\{.\}$ denotes fractional part of x. Now answer the following.

28. For $h(x)$ which one of the following is true.

A) $h(x)$ is continuous at $x=0$ and $x=\frac{\pi}{2}$

B) $h(x)$ is continuous at $x=0$ but discontinuous at $x=\frac{\pi}{2}$

C) $h(x)$ is discontinuous at $x=0$ but continuous at $x=\frac{\pi}{2}$

D) $h(x)$ is discontinuous at $x=0$ & $x=\frac{\pi}{2}$

29. For $f(x)$, which of the following is true

A) $f(x)$ is continuous at $x=0$ and $x=\frac{\pi}{2}$

B) $f(x)$ is continuous at $x=0$ but discontinuous at $x=\frac{\pi}{2}$

C) $f(x)$ is discontinuous at $x=0$ but continuous at $x=\frac{\pi}{2}$

D) $f(x)$ is discontinuous at $x=0$ & $x=\frac{\pi}{2}$

30. $g(\pi/4)+g(2\pi/3)+g(0)+g(\pi/2)=$

A) 0 B) 1 C) -1 D) 2

PASSAGE : 3

Let $f(x)=\begin{cases}x+2, & 0\le x<2\\ 6-x, & x\ge 2\end{cases}$,

$$g(x)=\begin{cases}1+\tan x, & 0\le x<\frac{\pi}{4}\\ 3-\cot x, & \frac{\pi}{4}\le 4<\pi\end{cases}$$

31. $f(g(x))$ **is**

A) discontinuous at $x=\frac{\pi}{4}$

B) differentiable at $x=\frac{\pi}{4}$

C) continuous but non-differentiable at $x=\frac{\pi}{4}$

D) differentiable at $x=\frac{\pi}{4}$, but derivative is not continuous.

32. The number of points on non-differentiability of $h(x)=|f(g(x))|$ is

A) 1 B) 2 C) 3 D) 4

33. The range of $h(x)=f(g(x))$ is

A) $(-\infty,\infty)$ B) $(4,\infty)$

C) $(-\infty,4)$ D) [-4, 5]

PASSAGE : 4

Let $f(x)=2+|x-1|$ and

$g(x)=\min(f(t))$ where $x\le t\le x^2+x+1$

then answer the following:

34. Number of points of discontinuity of $g(x)$ is

A) 1 B) 2 C) 3 D) 0

35. Number of points where the function $g(x)$ is not differentiable is

A) 1 B) 2 C) 3 D) 0

36. Range of $g(x)$ is

A) $[2,\infty)$ B) $[2,2)$ C) $[0,2]$ D) $(-\infty,\infty)$

PASSAGES : 5

Let A be a $n\times n$ matrix given by

$$A=\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{bmatrix}$$ **such that each horizontal row is an arithmetic progression and each vertical column is a geometrical progression. It is known that each column in geometric progression have the same common ratio. Given that $a_{24}=1$, $a_{42}=\frac{1}{8}$ and $a_{43}=\frac{3}{16}$.**

37. Let $S_n=\sum_{j=1}^{n} a_{4j}$, then $\lim_{n\to\infty}\frac{S_n}{n^2}$ is equal to

A) $\frac{1}{4}$ B) $\frac{1}{8}$ C) $\frac{1}{16}$ D) $\frac{1}{32}$

38. Let d_i be the common difference of the elements in i^th row, then $\sum_{i=1}^{n} d_i$

A) n B) $\frac{1}{2}-\frac{1}{2^{n+1}}$

C) $1-\frac{1}{2^n}$ D) $\frac{n+1}{2^n}$

39. The value of $\lim_{n\to\infty}\sum_{i=1}^{n} a_{ii}$ is equal to

A) $\frac{1}{4}$ B) $\frac{1}{2}$ C) 1 D) 2

PASSAGE : 6

If $f(x)=x^2-2|x|$

$$g(x)=\begin{cases} \text{minimum}\{f(t):-2\le t\le x\}, & -2\le x<0 \\ \text{minimum}\{f(t):0\le t\le x\}, & 0\le x\le 3 \end{cases}$$

then

40. The function $y=|f(x)|$ is differentiable for

A) $x\in R$ B) $x\in R-\{0\}$

C) $x\in R-\{0,2\}$ D) (-2, 2)

41. The function $g(x)$ is differentiable for

A) $x\in[-2,3]$

B) $x\in[-2,3]-\{-1,0,1\}$

C) $x\in[-2,3]-\{0,1\}$

D) $x\in[-2,3]-\{-1,0,2\}$

42. $g(x)=$ ..... $(x\in[-1,0]$

A) 0 B) $\frac{-1}{2}$ C) -1 D) 7 / 3

PASSAGE : 7

Suppose f, g and h be three real valued function defined on R.

Let $f(x)=2x+|x|$, $g(x)=\frac{1}{3}(2x-|x|)$ **and** $h(x)=f(g(x))$

43. The range of the function
$k(x)=1+\frac{1}{\pi}\left(\cos^{-1}(h(x))+\cot^{-1}(h(x))\right)$ **is equal to**

A) $\left[\frac{1}{4},\frac{7}{4}\right]$ B) $\left[\frac{5}{4},\frac{11}{4}\right]$ C) $\left[\frac{1}{4},\frac{5}{4}\right]$ D) $\left[\frac{7}{4},\frac{11}{4}\right]$

44. The domain of definition of the function
$l(x)=\sin^{-1}(f(x)-g(x))$ **is equal to**

A) $\left[\frac{3}{8},\infty\right)$ B) $(-\infty,1]$
C) $[-1,1]$ D) $\left(-\infty,\frac{3}{8}\right]$

45. The function
$T(x)=f(g(f(x)))+g(f(g(x)))$ **is**
A) continuous and diferentiable in $(-\infty,\infty)$
B) continuous but not derivable $\forall x\in R$
C) neither continuous nor derivable $\forall x\in R$
D) an odd function

PASSAGES : 8

Consider the function $f:R\to R$**, defined as** $f(x)=\begin{cases}x^2-x+3, & x\in(-\infty,3)\cap Q\\ x+a, & x\in(-\infty,2)-Q\\ 2^x+1, & x\in(2,3)-Q\\ 9\tan\left(\frac{\pi x}{12}\right), & x\in[3,6)\end{cases}$

46. If $f(x)$ **is continuous at x = 2 then the vlaue of a is**
A) 1 B) 2
C) 3 D)indeterminate

47. The function $f(x)$ **at x = 3**
A) has non-removable discontinuity
B) has removable discontinuity
C) is differentiable
D) is continuous but not differentiable

48. $f'(4)$ **is equal to**
A) π B) 3π C) $\frac{3\pi}{2}$ D) $\frac{3\pi}{16}$

MATRIX MATCHING QUESTIONS

49. <u>Column I</u>

A) $f(x)=\begin{cases}\frac{1}{|x|} & for\ |x|\ge 1\\ ax^2+b & for\ |x|<1\end{cases}$ is differentiable every where and $|k|=a+b$, then value of k is

B) If $f(x)=\operatorname{sgn}(x^2-ax+1)$ has exactly one point of discontinuity, then the value of a can be

C) $f(x)=[2+3|n|\sin x], n\in N, x\in(0,\pi)$ has exactly 11 points of discontinuity, then the value of n is

D) $f(x)=||x|-2+a|$ has exactly three points of non-differentiability, then the value of a is

<u>Column II</u>
P) 2
Q) –2
R) 1
S) –1

50. Match the following
<u>Column I</u>

A) If $f(x)=\frac{1-\cos(1-\cos x)}{x^4}$ **is continuous every where** $f(0)=\frac{1}{\lambda}$ **and let** μ **be he number of points** $g(x)=\max.(x^2-1,4,x^2+1)$ **is non-differentiable , then** $\lambda+\mu$ **is divisible by**

B) ' f **' be a continuous function on R. If** $f\left(\frac{1}{3^n}\right)=(\cos e^n)3^{-n^2}+\frac{n^2}{n^2+n+1}$ **and** $f(0)=\frac{1}{\lambda}$ **and let** μ **be the number of points if** $g(x)=\begin{cases}\min\{x,x^2\}, & x\ge 0\\ \min.\{2x,x^2-1,x<0\}\end{cases}$ **is**

non differentiable in $[-2,2]$, then $\lambda+\mu$ is divisible by

C) If $f(x)=\dfrac{\sqrt{2}\cos x-1}{\cot x-1}$ is continuous at $x=\dfrac{\pi}{4}$ and $f\left(\dfrac{\pi}{4}\right)=\dfrac{1}{\lambda}$ and μ be the jump of the function at the point of discontinuity of the function $g(x)=\dfrac{1-k^{1/x}}{1+k^{1/x}}(k>0)$, then $|\lambda-\mu|$ is divisible by

Column II

P) 2
Q) 3
R) 4
S) 5

51. **Let $f(x)$ be a real valued function defined by $f(x)=x^2-2|x|$ and**

$$g(x)=\begin{cases}\min imum\{f(t):-2\le t\le x\}, & x\in[-2,0)\\ \max imum\{f(t):0\le t\le x\}, & x\in[0,3]\end{cases}$$

Column I

A) $g(x)$ is not continuous at x equal to

B) $g(x)$ is not derivable at x equal to

C) Number of integral critical points of $g(x)$ is equal to

D) Absolute maximum value of $g(x)$ is equal to

Column II

P) –2
Q) 0
R) 1
S) 2
T) 3

52. $L=\lim\limits_{x\to0}\left[\dfrac{n_1\sin x}{x}\right]$, $M=\lim\limits_{x\to0}\left[\dfrac{n_2x}{\sin x}\right]$

$N=\lim\limits_{x\to0}\left[\dfrac{n_3\tan x}{x}\right]$ **and** $P=\lim\limits_{x\to0}\left[\dfrac{n_4x}{\tan x}\right]$

where [.] denotes the greatest integer function and $n_1,n_2,n_3,n_4\in N$.

Column - I

A) For $n_1=3,n_4=9$ then P/L is divisible by

B) For $n_2=12,n_3=2$, then M/N is dibisible by

C) For $n_2=3,n_4=10$ then P+M is divisible by

D) For $n_1=4,n_2=5,n_3=6,n_4=7$ then L+N+P+M is dibisible by

Column - II

P) 2
Q) 3
R) 4
S) 5
T) 6

INTEGER QUESTIONS

53. **If $\lim\limits_{x\to0}\dfrac{1-\cos\left(1-\cos\dfrac{x}{2}\right)}{2^m x^n}$ is equal to the left hand derivative of $e^{-|x|}$ at x = 0, then find the value of (n^2+m).**

54. **If $f(x)=\begin{cases}\operatorname{sgn}(x-2)\times[\log_e x], & 1\le x\le3\\ \{x^2\}, & 3<x\le3.5\end{cases}$, where $[\bullet]$ denotes the greatest integer function and $\{\bullet\}$ represents the fractional part function, then the number of points of discontinuity is.................**

55. **Let $\lim\limits_{x\to1}\left[\dfrac{x^a-ax+a-1}{(x-1)^2}\right]=f(a)$. Then the value of $f(4)$ is ....**

56. **If f and g are two differentiable functions with $g'(a)=2,\ g(a)=b$, such that fog = identity function, then $2f'(b)$ is**

57. **If $f(x),g(x)$ be twice differentiable function on [0,2] satisfying**

$f''(x)=g''(x), f'(1)=2, g'(1)=4$ and $f(2)=3, g(2)=9$, then –f(x) + g(x) at x = 2 equals …………..

58. Let S denote sum of the series $\frac{3}{2^3}+\frac{4}{2^4.3}+\frac{5}{2^6.3}+\frac{6}{2^7.5}+..........\infty$.
Then the value of S^{-1} is

59. The largest value of the non-negative integer a for which

$$\lim_{x\to1}\left\{\frac{-ax+\sin(x-1)+a}{x+\sin(x-1)-1}\right\}^{\frac{1-x}{1-\sqrt{x}}}=\frac{1}{4}$$ is

60. The value of

$$\left[\lim_{n\to\infty}\left(2\times3^2\times2^3\times3^4.....2^{n-1}\times3^n\right)^{\frac{1}{(n^2+1)}}\right]^4$$ is

61. If $L=\lim_{x\to0}\frac{\sin x^4-x^4\cos x^4+x^{20}}{x^4\left(e^{2x^4}-1-2x^4\right)}$, then the value of 1/L is ___________

62. If $\lim_{x\to0}\sin\left(\frac{\pi\left(1-\cos^m x\right)}{x^n}\right)$ exists, where $m,n\in N$, then the sum of all possible values of *n* is ___________.

63. If $\lim_{x\to\infty}\left(e^{2x}+e^x+x\right)^{\frac{1}{x}}=e^a$, then the value of a is

64. If a and b are the numbers of points of non differentiability of $f(x)=\left[\sin^{-1}x\right]$ and $f(x)=\left[\frac{2}{1+x^2}\right]$, $x\ge0$, (where [.] represents greatest integer function) respectively, then the value of $a+b$ is

65. The least integral value of a for which the function $f(x)=\left[(x-2)^3/a\right]\sin(x-2)+a\cos(x-2)$, where [.] denotes the greatest integer function, is continuous in [0,2] is _____

66. If a is the number of points of continuity of $f(x)=\begin{cases}x-1, & x \text{ is rational}\\ x^2-x-2, & x \text{ is irrational}\end{cases}$,
b is the number of points of discontinuity of $f(x)=\operatorname{sgn}\left(x^3-3x+1\right)$, *c* is the number of points of non-differentiability of $f(x)=(\log x)\left|x^2-4x+3\right|+2(x-2)^{1/3}$, and *d* is the number of points where the graph of $f(x)=(\log x)\left|x^2-4x+3\right|+2(x-2)^{1/3}$ has a sharp turn then the value of $a+b+c+d$ is _____.

67. The number of points of discontinuity of $f(x)=\lim_{n\to\infty}\left[\left(x^{2n}-1\right)/\left(x^{2n}+1\right)\right]$ is ______

68. Given the continuous function $f(x)=\begin{cases}2x+3 & x\le1\\ -x^2+6 & x>1\end{cases}$, then numner of points where $\left|f(|x|)\right|$ is non-differentiable is __________.

KEY - LEVEL - VI

SINGLE ANSWER QUESTIONS

1. A	2. A	3. C	4. C
5. C	6. C	7. C	8. D
9. D	10. B	11. A	12. B
13. D	14. B	15. D	

MULTI ANSWER QUESTIONS

16. A,C	17. B, D
18. A, C	19. C, D
20. A, D	21. A, B, C, D
22. A, B	23. A, C, D
24. A,B,D	

COMPREHENSION QUESTIONS

25. C	26. B	27. A	28. B
29. D	30. A	31. C	32. B
33. C	34. C	35. C	36. A

37. D	**38. C**	**39. D**	**40. B**
41. C	**42. C**	**43. B**	**44. D**
45. B	**46-C**	**47-D**	**48. B**

MATRIX MATCHING QUESTIONS

49. A-R, S; B-P, Q; C-P, Q; D-P, R
50. A-PS; B-PR,C-PQRS
51. A-Q, B-Q, S, C-T, D-T
52. A-PR; B-PQT; C-P,Q,R,T; D-P,S

INTEGER QUESTIONS

53. 9	**54. 4**	**55. 6**	**56. 1**
57. 6	**58. 2**	**59. 2**	**60.6**
61. 6	**62. 3**	**63. 2**	**64. 5**
65. 9	**66. 8**	**67. 2**	**68. 5**

HINTS-LEVEL - VI

SINGLE ANSWER QUESTIONS

1. $x^2+2x+3+\sin\pi x=(x+1)^2+2+\sin\pi x>1$

$\therefore\ f(x)=1\ \forall\ x\in R$

2. If $g(x)=(\sin x)^{\ln x}$

$$f(x)=g'(x)=(\sin x)^{\ln x}\left[\cot x(\ln x)+\frac{\ln(\sin x)}{x}\right]$$

Hence $f\left(\frac{\pi}{2}\right)=g'\left(\frac{\pi}{2}\right)=1(0+0)=0$ (Ans.)

3. $$g'(0)=b=\lim_{x\to 0}\frac{x^2+x\tan x-x\tan 2x}{x(ax+\tan x-\tan 3x)}=\lim_{x\to 0}\frac{x+\tan x-\tan 2x}{ax+\tan x-\tan 3x}$$

$$=\lim_{x\to 0}\frac{x+\left(x+\frac{x^3}{3}+\frac{2}{15}x^5+.....\infty\right)-\left(2x+\frac{8x^3}{3}+\frac{2}{15}.32x^5+.....\infty\right)}{ax+\left(x+\frac{x^3}{3}+\frac{2}{15}x^5+.....\infty\right)-\left(3x+\frac{27x^3}{3}+\frac{2}{15}.243x^5+.....\infty\right)}$$

On simplifying a = 2, b= $\frac{7}{26}$.

4. We have $\underset{n\to\infty}{Lim}\frac{n.3^n}{n(x-2)^n+n.3^{n+1}-3^n}=\frac{1}{3}$;

so $\underset{n\to\infty}{Lim}\frac{1}{\left(\frac{x-2}{3}\right)^n+3-\frac{1}{n}}=\frac{1}{3}$

Clearly $-1<\frac{x-2}{3}<1\Rightarrow\quad -1<x<5$

$\therefore$ Possible integers in the range 'x' are 0,1,2,3,4 $\Rightarrow$ 5 integers.

5. For continuity of f at x = 0, we have

$$k=f(0)=\lim_{x\to 0^+}f(x)=\lim_{x\to 0}\frac{e^{\tan x}-e^x}{\tan x-x}+\lim_{x\to 0}\frac{\ln(\sec x+\tan x)-x}{\left(\frac{\tan x-x}{x^3}\right)x^3}$$

$$=\lim_{x\to 0}\frac{e^x\left(e^{\tan x-x}-1\right)}{\tan x-x}+3\lim_{x\to 0}\frac{\ln(\sec x-\tan x)-x}{x^3}$$

$$=1+3\lim_{x\to 0}\frac{\sec x-1}{3x^2}\quad \text{(Using LH Rule)}$$

$$=1+\frac{1}{2}=\frac{3}{2}$$

6. $$\underset{x\to 0^+}{Lt}f(x)=\underset{h\to 0}{Lt}\frac{\sin^{-1}(1-h)}{\sqrt{2}(1-h)(1+h)}.\underset{h\to 0}{Lt}\frac{\cos^{-1}(1-h^2)}{h}=\frac{\pi}{2}.$$

$$\underset{x\to 0^-}{Lt}f(x)=\underset{h\to 0}{Lt}\frac{\cos^{-1}h(2-h).\sin^{-1}h}{\sqrt{2}(1-h)(2-h)}=\frac{\pi}{4\sqrt{2}}.$$

7. $$\left(6^{\frac{1}{\cos^2 x}}\right)^{2\cos^2 x}.\underset{x\to\frac{\pi}{2}}{\text{Lim}}\left(\left(\frac{1}{3}\right)^{\frac{1}{\cos^2 x}}+\left(\frac{1}{2}\right)^{\frac{1}{\cos^2 x}}+\left(\frac{2}{3}\right)^{\frac{1}{\cos^2 x}}+\left(\frac{5}{6}\right)^{\frac{1}{\cos^2 x}}+1\right)=36$$

8. $:\because 2<f(x)<\sqrt{26}$

$\therefore\frac{2}{c}<\frac{f(x)}{c}<\frac{\sqrt{26}}{c}$

since, $g(x)=\left[\frac{f(x)}{c}\right]$ is continuous $\forall x\in R$

$\Rightarrow g(x)=0$ for $c=6$

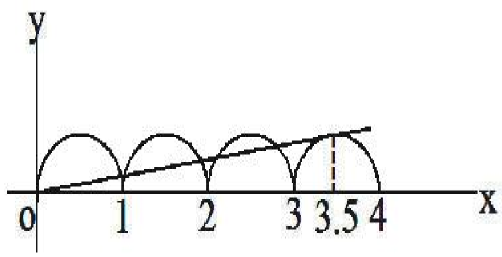
9. We have $\lim\limits_{x\to 1^-}f(x)=\lim\limits_{h\to 0}f(1-h)$

$$= \lim_{h\to 0}\frac{\log\left(4+h^2\right)}{\log\left(1-4h\right)} = -\infty \quad \text{and}$$

$$\lim_{x\to 1^+} f(x) = \lim_{h\to 0} f(1+h) = \lim_{h\to 0}\frac{\log\left(4+h^2\right)}{\log\left(1+4h\right)} = \infty$$

So, $f\left(1^-\right)$ and $f\left(1^+\right)$ do not exist.

10. $f(x) = \max\left\{\frac{x}{n}, |\sin \pi x|\right\}$



Thus, for the maximum points of non differentiability, graphs of $y = \frac{x}{7}$ and $y = |\sin \pi x|$ must intersect at maximum number of points which occurs when $n > 3.5$.
Hence, the least value of n is 4.

11. $\lim_{x\to\infty}\frac{\log_{10}(x+2)+[x]+2}{[x]2}$

$$= 1 + \lim_{x\to\infty}\frac{\log_{10}(x+2)}{[x]+2}$$

$$= 1 + \lim_{x\to\infty}\frac{\frac{\log_{10}(x+2)}{x}}{\frac{[x]}{x}+\frac{2}{x}}$$

$$= 1 + 0$$
$$= 1$$

12. Let $\frac{y}{x} = a \Rightarrow x + y = \frac{-3}{2a}, x + y = \frac{1}{2} - a$

$$-\frac{3}{2a} = \frac{1}{2} - a \Rightarrow -\frac{3}{2a} = \frac{1-2a}{2}$$

$$\Rightarrow -3 = a - 2a^2 \Rightarrow 2a^2 - a - 3 = 0$$

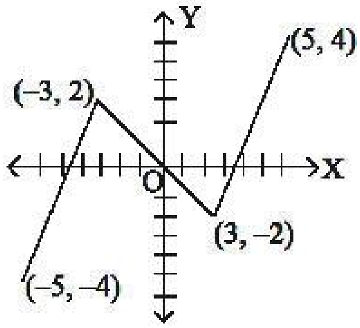
$$\Rightarrow a = -1 \text{ or } a = \frac{3}{2} \qquad \therefore \frac{x}{y} = \frac{2}{3}$$

$$\therefore \lim_{n\to\infty}\sum_{r=0}^{n}\left(\frac{x}{y}\right)^r = 3$$

13. $f(x) + f(-x) = 0 \Rightarrow f(x)$ is an odd function. Since points $(-3,2)$ and $(5,4)$ lie on the curve

$\therefore$ $(3,-2)$ and $(-5,-4)$ will also lie on the curve.

For minimum number of roots, graph of continuous function $f(x)$ is as follows.



from the above two graphs of $f(x)$ it is clear that equation $f(x) = 0$ has atleast three real roots.

14. Given $\left(1+\frac{1}{n}\right)^{n+x_n} = e$

taking log

$$(n+x_n)\ln\left(1+\frac{1}{n}\right) = 1 \Rightarrow n + x_n = \frac{1}{\ln\left(1+\frac{1}{n}\right)}$$

$$\Rightarrow x_n = \frac{1}{\ln\left(1+\frac{1}{n}\right)} - n \qquad .....(1)$$

let $\frac{n+1}{n} = u \Rightarrow nu = n+1 \Rightarrow n = \frac{1}{u-1}$

$$x_n = \lim_{u\to 1}\left(\frac{1}{\ln u} - \frac{1}{u-1}\right) = \lim_{u\to 1}\frac{(u-1)-\ln u}{(u-1)\ln u}$$

$$= \underset{u\to 1}{Lim}\frac{1-\frac{1}{u}}{\frac{u-1}{u}+\ln u} = \underset{u\to 1}{Lim}\frac{\frac{1}{u^2}}{\frac{1}{u^2}+\frac{1}{u}} = \frac{1}{2}$$

15. $\cos^3 x - \cos^2 x \to 0$ from LHS

$\sin^2 x - \sin^3 x \to 0$ from RHS

$\therefore$ the given line reduces to $4y = 3x$

MULTI ANSWER TYPE QUESTIONS

16. a) $\underset{h\to 0}{\text{Lim}}\frac{f(c+h)-f(c)+f(c)-f(c-h)}{2h}$

$$=\frac{1}{2}f^1(c)+\frac{1}{2}f^1(c)=f^1(c)$$

b) existence of limit cannot confirm differentiablility

17. (b,c,d)

(a) $f(0^-) = \lim_{h\to 0} f(o-h) = \lim_{h\to 0}\frac{1}{1+2^{\cot(o-h)}}$

$$= \lim_{h\to 0}\frac{1}{1+2^{-\coth}} = \frac{1}{1+2^{-\infty}} = \frac{1}{1+0} = 1$$

$\because f(0^-) \neq f(0^+)$

(b)

$$f(0^-) = \lim_{h\to 0} f(0-h) = \lim_{h\to 0}\cos\left(\frac{|\sin(0-h)|}{0-h}\right)$$

$$= \lim_{h\to 0}\cos\left(\frac{\sinh}{-h}\right)$$

$$= \cos\left(-\lim_{h\to 0}\cos\frac{\sinh}{-h}\right) = \cos(-1) = \cos 1$$

and

$$f(0^+) = \lim_{h\to 0} f|0+h| = \lim_{h\to 0}\cos\left(\frac{|\sin(0-h)|}{0-h}\right)$$

$$= \lim_{h\to 0}\cos\left(\frac{\sinh}{h}\right) = \cos\left(\lim_{h\to 0}\frac{\sinh}{h}\right) = \cos 1$$

$\therefore \quad f(0^-) = f(0^+) \neq f(0)$

(C) : $f(0^-) = \lim_{h\to 0} f(0-h) = \lim_{h\to 0}(-h)\sin\left(\frac{\pi}{-h}\right)$

$$= \lim_{h\to 0} hh\sin\left(\frac{\pi}{h}\right)$$

$$= 0\times\sin\infty$$

$$= 0\times(\text{lies between - 1+to 1})$$

and $f(0^+) = \lim_{h\to 0} f(0+h) = \lim_{h\to 0} h\sin\left(\frac{\pi}{h}\right)$

$$= 0\times\sin\infty = 0\times(\text{lies between -1 to 1}) = 0$$

$\therefore f(0^-) = f(0^+) \neq f(0)$

(d) : $f(0^-) = \lim_{h\to 0} f(0-h) = \lim_{h\to 0}\frac{1}{\ln|0-h|}$

$$= \lim_{h\to 0}\frac{1}{\ln h}$$

$$= \frac{1}{\ln 0} = \frac{1}{-\infty} = 0$$

and $f(0^+) = \lim_{h\to 0} f(0+h) = \lim_{h\to 0}\frac{1}{\ln|0+h|} = \lim_{h\to 0}\frac{1}{\ln h}$

$$= \frac{1}{-\infty} = 0$$

$\therefore \quad f(0^-) = f(0^+) \neq f(0)$

18. We have, $f(x) = \min\{1, \cos x, 1-\sin x\}$

$\therefore$ $f(x)$ can be rewritten as

$$f(x) = \begin{cases} \cos x, & -\frac{\pi}{2} \le x \le 0 \\ 1-\sin x, & 0 < x \le \frac{\pi}{2} \\ \cos x, & \frac{\pi}{2} < x \le \pi \end{cases}$$

$$\Rightarrow f'(x) = \begin{cases} -\sin x, & -\frac{\pi}{2} \le x \le 0 \\ -\cos x, & 0 < x \le \frac{\pi}{2} \\ -\sin x, & \frac{\pi}{2} < x \le \pi \end{cases}$$

$\because f'(0) = 0$

hence, f(x) has local maxima at 0 and f(x) is not differentiable at $x = 0$.

19. f(x) is discontinuous at $\frac{\pi}{2}, \pi\frac{3\pi}{2}, 2\pi, ...$

and $f(x)$ is periodic with period π

Range of $f(x)$ is $(-\infty,-1]\cup[0,1]$

20. $f(x)=\begin{cases} e^x & x<0 \\ 2-e^x & 0\le x<\ln 2 \\ e^x-2 & x\ge \ln 2\end{cases}$

f is continuous $\forall x\in R$, but is not differentiable at $x=0,\ln 2$

21. We have $\lim\limits_{k\to 0^+} f(x)=\lim\limits_{k\to 0^+}\dfrac{\tan^2\{x\}}{\left(x^2-[x]^2\right)}$

$=\lim\limits_{x\to 0^+}\dfrac{\tan^2 x}{x^2}=1$ .....(1)

$\left[\because x\to 0^+;[x]=0=\{x\}=x\right]$

Also $\lim\limits_{k\to 0^-} f(x)=\lim\limits_{x\to 0^-}\sqrt{\{x\}\cot\{x\}}=\sqrt{\cot 1}$ .....(2)

$\left[\because x\to 0^-;[x]=-1\Rightarrow\{x\}=x+1\Rightarrow\{x\}\to 1\right]$

Also, $\cot^{-1}\left(\lim\limits_{x\to 0^-} f(x)\right)^2=\cot^{-1}(\cot 1)=1$

22. For maximum points of discontinuity of $f(x)=\operatorname{sgn}\left(x^2-ax+1\right)$, $x^2-ax+1=0$ must have two distinct roots, for which $D=a^2-4>0\Rightarrow a\in(-\infty,-2)\cup(2,\infty)$.

23. For continuity at x =1

$\lim\limits_{x\to 1^+} f(x)=\lim\limits_{x\to 1^+}\left(x^2\operatorname{sgn}[x]+\{x\}\right)=1+0=1$

$\lim\limits_{x\to 1^-} f(x)=\lim\limits_{x\to 1^-}\left(x^2\operatorname{sgn}[x]+\{x\}\right)=1\operatorname{sgn}(0)+1=1$

Also, $f(1)=1$

$\therefore$ L.H.L = R.H.L = $f(1)$.

Hence $f(x)$ is continuous at x =1.

Now for differentiability,

$f'(1^+)=\lim\limits_{h\to 0}\dfrac{f(1+h)-f(1)}{h}$

$=\lim\limits_{h\to 0}\dfrac{(1+h)^2\operatorname{sgn}[1+h]+\{1+h\}-1}{h}$

$=\lim\limits_{h\to 0}\dfrac{(1+h)^2+h-1}{h}=\lim\limits_{h\to 0}\dfrac{h^2+3h}{h}=3$

and $f'(1^{-1})=\lim\limits_{h\to 0}\dfrac{f(1-h)-f(1)}{-h}$

$=\lim\limits_{h\to 0}\dfrac{(1-h)^2\operatorname{sgn}[1-h]+\{1-h\}-1}{-h}$

$=\lim\limits_{h\to 0}\dfrac{(1-h)^2+1-h-1}{-h}=\lim\limits_{h\to 0}\dfrac{h^2-3h}{-h}=3$

$f'(1^+)=f'(1^-)$

Hence, $f(x)$ is differentiable at x =1.

Now at x =2,

$\lim\limits_{x\to 2^-} f(x)=\lim\limits_{x\to 2^-}\left(x^2\operatorname{sgn}[x]+\{x\}\right)=4\times 0+1$

$\lim\limits_{h\to 2^+} f(x)=\lim\limits_{h\to 2^+}\left(\sin x+|x-3|\right)=1+\sin 2$

Hence $LHL\ne RHL$

Hence, $f(x)$ is discontinuous at x =2 and then $f(x)$ is also non differentiable at x = 2.

24. $f(x)=\sum\limits_{r=1}^{n}\left(x^{2r}+\dfrac{1}{x^{2r}}+2\right)=\sum\limits_{r=1}^{n}x^{2r}+\sum\limits_{r=1}^{n}\dfrac{1}{x^{2r}}+2n$

$=\left(x^2+x^4+......+x^{2n}\right)+\left(\dfrac{1}{x^2}+\dfrac{1}{x^4}+.....+\dfrac{1}{x^{2n}}\right)+2n$

$=\dfrac{x^2\left(1-x^{2n}\right)}{\left(1-x^2\right)}+\dfrac{1}{x^2}\dfrac{\left(1-\dfrac{1}{x^{2n}}\right)}{1-\dfrac{1}{x^2}}+2n$

$f(x)=\dfrac{x^2\left(1-x^{2n}\right)}{1-x^2}+\dfrac{\left(1-x^{2n}\right)}{\left(1-x^2\right)x^{2n}}+2n$

$$f(x)=\frac{(1-x^{2n})}{1-x^2}\left(x^2+\frac{1}{x^{2n}}\right)+2n \quad x\neq\pm1$$

$$\therefore\ (f(x)-2n)(1-x^2)=(1-x^{2n})\left(x^2+\frac{1}{x^{2n}}\right)$$

now consider

$$g(x)=\lim_{n\to\infty}\left((f(x)-2n)x^{-2n-2}(1-x^2)\right) \text{ for}$$

$x\neq\pm1$

$$=\lim_{n\to\infty}(1-x^{2n})\left(x^2+\frac{1}{x^{2n}}\right).\frac{1}{x^{2n+2}};\ x\neq\pm1$$

$$=\lim_{n\to\infty}\frac{(1-x^{2n})(x^{2n+2}+1)}{(x^{2n})(x^{2n+2})}$$

$$=\lim_{n\to\infty}\left(-1+\frac{1}{x^{2n}}\right)\left(1+\frac{1}{x^{2n+2}}\right)$$

now $g(x)=\begin{cases}-1 & if\ |x|>1\\ -\infty & if\ |x|<1\\ 0 & if\ x=\pm1\end{cases}$

now clearly g has non removable infinite type of discontinuity at x = 1 and –1
g is continuous at x = 2

COMPREHENSION QUESTIONS

PASSAGE : 1 25 -27

$\because (gof)(x)$ is continuous $g(x)$ and $f(x)$ are also continuous , $\underset{x\to0^-}{Lt} f(x)=f(0)$

$\underset{h\to0}{Lt}(0-h)+a=1 \Rightarrow a=1$

$\underset{x\to0^-}{Lt} g(x)=g(0)$, $\underset{h\to0}{Lt} 0-h+1=(0-1)^2+b$

$\Rightarrow 1=1+b \ \Rightarrow b=0$

$$(gof)(x)=g(f(x))=\begin{cases}f(x)+1, & f(x)<0\\ (f(x)-1)^2, & f(x)\geq0\end{cases}$$

$$=\begin{cases}x+2, & x<-1\\ x^2, & -1\leq x<1\\ (x-2)^2, & x\geq1\end{cases}$$

$\therefore\ (gof)(x)$ is even for all $x\in(-1,1)$

PASSAGE : 2 Q.No. 28 to 30

Conceptual

PASSAGE : 3 Q.No. 31 to 33

For $0\leq x<\frac{\pi}{4}$, $g(x)=1+\tan x$

$x\in\left[0,\frac{\pi}{4}\right)\Rightarrow 1+\tan x\in[1,2)$

So $f(g(x))=f(1+\tan x)=1+\tan x+2$

and for $x\in\left[\frac{\pi}{4},\pi\right), g(x)=3-\cot x$

$x\in\left[\frac{\pi}{4},\pi\right)\Rightarrow 3-\cot x\in[2,\infty)$

So $f(g(x))=f(3-\cot x)=6-(3-\cot x)$

Let

$$h(x)=f(g(x))=\begin{cases}3+\tan x, & 0\leq x<\frac{\pi}{4}\\ 3+\cot x, & \frac{\pi}{4}\leq x<\pi\end{cases}$$

Clearly, $f(g(x))$ is continuous in $[0,\pi)$

Now $h'\left(\frac{\pi^+}{4}\right)=\lim_{x\to\frac{\pi}{4}^+}(-\operatorname{cosec}^2 x)=-2$

$h'\left(\frac{\pi^-}{4}\right)=\lim_{x\to\frac{\pi}{4}^-}(\sec^2 x)=2$

So $f(g(x))$ is differentiable everywhere in $[0,\pi)$ other than at $x=\frac{\pi}{4}$.

$$|f(g(x))|=\begin{cases}|3+\tan x|, & 0\leq x<\frac{\pi}{4}\\ |3+\cot x|, & \frac{\pi}{4}\leq x<\pi\end{cases}$$

Which is non differentiable at $x=\frac{\pi}{4}$ and where $3+\cot x=0$ or $x=\cot^{-1}(-3)$

For $x \in \left[0, \frac{\pi}{4}\right), 3 + \tan x \in [3, 4)$

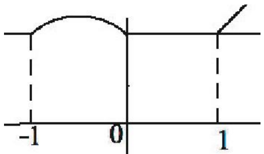
For $x \in \left[\frac{\pi}{4}, \pi\right), 3 + \cot x \in (-\infty, 4]$

Hence, the range is $(-\infty, 4]$.

PASSAGE : 4 Q.No. 34 to 36

$$f(t) = \begin{cases} 3 - t, & t \le 1 \\ t + 1, & t > 1 \end{cases}$$

$$g(x) = \begin{cases} 3 - (x^2 + x + 1) & x \in [-1, 0) \\ 2 & x \in (-\infty, -1) \cup (0, 1] \\ x + 1 & x \in [1, \infty) \end{cases}$$



From graph we can easily give the answer

PASSAGES : 5 (Q.No. 37 to 39)

$a_{43} = a_{42} + d_4 \Rightarrow d_4 = \frac{1}{16}$

(common difference of 4th row)

$a_{41} = a_{42} - d_4 = \frac{1}{16}$

$\therefore\ a_{41} = \frac{1}{16},\ a_{42} = \frac{2}{16},\ a_{43} = \frac{3}{16},\ a_{44} = \frac{4}{16},$

$......., a_{4n} = \frac{n}{16}$

now all elements of 4th row are known

$S_n = \sum_{j=1}^{n} a_{4j} = \frac{n(n+1)}{2(16)};$

$\underset{n\to\infty}{Lim} \frac{S_n}{n^2} = \frac{1}{32}$ (Ans: D)

$a_{24} r^2 = a_{44} = \frac{4}{16} \Rightarrow r^2 = \frac{1}{4} \Rightarrow r = \frac{1}{2}$

(common ratio for all G.P. is $\frac{1}{2}$)

note that all the lements of 4th row are known

and common ratio $\frac{1}{2}$ is also known therefore

$(m \times n)$ matrix is constructed.

$$A = \begin{bmatrix} \frac{1}{2} & \frac{2}{2} & \frac{3}{2} & \frac{4}{2} & \cdots & \frac{n}{2} \\ \frac{1}{2^2} & \frac{2}{2^2} & \frac{3}{2^2} & \frac{4}{2^2} & \cdots & \frac{n}{2^2} \\ \frac{1}{2^3} & \frac{2}{2^3} & \frac{3}{2^3} & \frac{4}{2^3} & \cdots & \frac{n}{2^3} \\ \frac{1}{2^4} & \frac{2}{2^4} & \frac{3}{2^4} & \frac{4}{2^4} & \cdots & \frac{n}{2^4} \\ \frac{1}{2^n} & \frac{2}{2^n} & \frac{3}{2^n} & \frac{4}{2^n} & \cdots & \frac{n}{2^n} \end{bmatrix}$$

Working: $a_{41} = \frac{1}{2^4}$, $a_{42} = \frac{2}{2^4}$ |||ly $a_{4i} = \frac{i}{2^4}$

now $\left.\begin{matrix} a_{11} \\ a_{21} \\ a_{31} \\ a_{41} \end{matrix}\right\}$ in G.P. $\quad a_{41} = a_{11}.r^3$

$\Rightarrow \frac{1}{16} = a_{11} \cdot \frac{1}{8}$

$\Rightarrow a_{11} = \frac{1}{2}$

again $a_{42} = \frac{2}{2^4}$

$\left.\begin{matrix} a_{12} \\ a_{22} \\ a_{33} \\ a_{42} \end{matrix}\right\}$ in G.P. $\quad a_{42} = a_{12}.r^3$

$\Rightarrow \frac{2}{16} = a_{12} \cdot \frac{1}{8}$

$\Rightarrow a_{12} = \frac{1}{2^2}$

$$\sum_{i=1}^{n} d_i = \sum_{i=1}^{n} \frac{1}{2^i} = \frac{\frac{1}{2}\left(1 - \frac{1}{2^n}\right)}{\left(1 - \frac{1}{2}\right)} = 1 - \frac{1}{2^n}$$

$$\sum_{i=1}^{n} a_{ii} = S = \frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \frac{4}{2^4} + ... + \frac{n}{2^n}$$

$$\frac{1}{2} S = \frac{1}{2^2} + \frac{2}{2^3} + \frac{3}{2^4} + ... + \frac{n-1}{2^n} + \frac{n}{2^{n+1}}$$

$$\frac{1}{2} S = \left(\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + ... + \frac{1}{2^n}\right) - \frac{n}{2^{n+1}}$$

$$S = 2\left(1 - \frac{1}{2^n}\right) - \frac{n}{2^n}$$

$$\underset{n\to\infty}{Lim} \sum_{i=1}^{n} a_{ii} = \underset{n\to\infty}{Lim}\left(2 - \frac{2}{2^n} - \frac{n}{2^n}\right)$$

$$= 2 - 0 - 0 \Rightarrow \underset{n\to\infty}{Lim} \sum_{i=1}^{n} a_{ii} = 2$$

PASSAGE : 6 (q.no. 40 to 42)

$$g(x)=\begin{matrix} f(x) & if & -2\le x\le -1 \\ -1 & if & -1\le x<0 \\ 0 & if & -\le x<1 \\ f(x) & if & 1\le x\le 3 \end{matrix}$$

PASSAGE : 7 (q.no. 43 to 45)

We have $f(x)=\begin{cases}3x, x\ge 0\\ x,\ \ x<0\end{cases}$ and

$$g(x)=\begin{cases}\frac{x}{3}, x\ge 0\\ x,\ \ x<0\end{cases}$$

Clearly f and g are inverse of each other

Now, $h(x)=f(g(x))=\begin{cases}3\left(\frac{x}{3}\right)=x,\ x\ge 0\\ x,\qquad x<0\end{cases}$

(i) As $h(x)=x\forall x\in R$

$\Rightarrow\ k(x)=1+\frac{1}{\pi}\left(\cos^{-1}x+cot^{-1}x\right)$

Domain of $k(x)=[-1,1]$ and $k(x)$ is decreasing function on $[-1,1]$

As $k(x)$ is continuous function on $[-1,1]$

Now, $k_{\min}(x=1)=1+\frac{1}{\pi}\left(\cos^{-1}1+\cot^{-1}1\right)$

$=1+\frac{1}{\pi}\left(0+\frac{\pi}{4}\right)=1+\frac{1}{4}=\frac{5}{4}$

$k_{\max}(x=-1)=1+\frac{1}{\pi}\left(\cos^{-1}(-1)+\cot^{-1}(-1)\right)$

$=1+\frac{1}{\pi}\left(\pi+\frac{3\pi}{4}\right)=1+\frac{7}{4}=\frac{11}{4}$

$\Rightarrow$ Range of $k(x)=\left[\frac{5}{4},\frac{11}{4}\right]$

(ii) We have

$f(x)-g(x)=(2x+|x|)-\frac{1}{3}(2x-|x|)$

$=\frac{4x}{3}+\frac{4}{3}|x|=\begin{cases}\frac{8}{3}x;\ x\ge 0\\ 0\ \ ;\ x<0\end{cases}$

$\therefore$ For domain of function,

$0\le\frac{8x}{3}\le 1\Rightarrow\ 0\le x\le\frac{3}{8}$

$\Rightarrow$ Domain of $l(x)=\left(-\infty,\frac{3}{8}\right]$

Note : Range of function $l(x)=\left[0,\frac{\pi}{2}\right]$

(iii) As f and g are ianverse of each other, so $T(x)=$

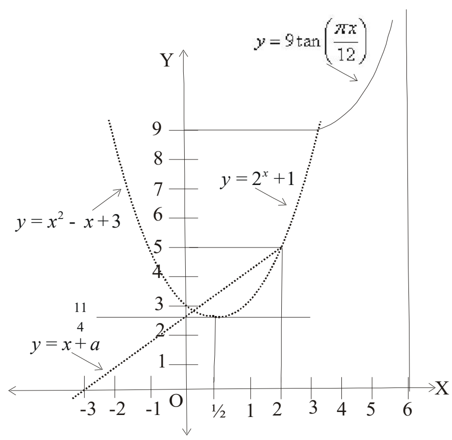
$T0x=f(g(f(x)))+g(f(g(x)))$

$=f(x)+g(x)=(2x+|x|)+\frac{1}{3}(2x-|x|)$

$\Rightarrow\ T(x)=\begin{cases}\frac{10x}{3}, x\ge 0\\ 2x,\ \ x<0\end{cases}$ Clearly, $T(x)$ is continuous but non-derivable at x $=0$

PASSAGES : 8 (Q.NO. 46 to 48)

$$f(x)=\begin{cases}x^2-x+3,\ if\ \ x\in Q\ \ in\ \ (-\infty,3)\\ x+a,\qquad if\ \ x\notin Q\ in\ (-\infty,2)\\ 2^x+1,\qquad if\ \ x\notin Q\ in\ (2,3)\\ 9\tan\left(\frac{\pi x}{12}\right), if\ x\ge 3\end{cases}$$



through irrational $\qquad f(2^-)=2+a$

through rational $\qquad f(2)=4-2+3=5$

Hence for continuity at x = 2

$2+a=5\Rightarrow a=3$

for $x \geq 3$, $f(x) = 9\tan\frac{\pi x}{12}$

$\Rightarrow f(3^+) = 9$; $f(3^-) = 2^3 + 1 = 9$

$\Rightarrow$ f is continuis at x = 3 obvious not derivable

$$f'(x) = 9\sec^2\left(\frac{\pi x}{12}\right)\frac{\pi}{12}$$

$$f'(4) = 9\sec^2\left(\frac{\pi}{3}\right)\frac{\pi}{12} = 3\pi$$

MATRIX MATCHING QUESTIONS

49. A)Thegiven function is clearly continuous at all points except possibly at $x = \pm 1$.

As $f(x)$ is an even function, so we need to check its continuity only at x =1.

$$\Rightarrow \lim_{x \to 1^-}\left(ax^2 + b\right) = \lim_{x \to 1^+}\frac{1}{|x|} \Rightarrow a = 1 = b = 1$$

Clearly, $f(x)$ is differentiable for all x, except possibly at $x = \pm 1$. As $f(x)$ is an even function, so we need to check its differentiability at x=1 onlyu.

$$\lim_{x \to 1^-}\frac{f(x) - f(1)}{x - 1} = \lim_{x \to 1^+}\frac{f(x) - f(1)}{x - 1}$$

$$\Rightarrow \lim_{x \to 1}\frac{ax^2 + b - 1}{x - 1} = \lim_{x \to 1}\frac{ax^2 + b - 1}{x - 1} = \lim_{x \to 1^+}\frac{\frac{1}{|x|} - 1}{x - 1}$$

$$\Rightarrow \lim_{x \to 1}\frac{ax^2 - a}{x - 1} = \lim_{x \to 1}\frac{-1}{x} \Rightarrow 2a = -1 \Rightarrow a = \frac{1}{2\frac{1}{2}}$$

Putting $a = -\frac{1}{2}$ in (1)

we get $b = \frac{3}{2} \Rightarrow |k| = 1 \Rightarrow k = \pm 1$

B) $f(x) = \operatorname{sgn}\left(x^2 - ax + 1\right)$ is discontinuous then $x^2 - ax + 1 = 0$ must have only one real root. Hence $a = \pm 2$.

C) $f(x) = \left[2 + 3|n|\sin x\right], n \in N$ has exactly 10 points of discontinuity in $x \in (0, \pi)$.

The required numnber of points are

$1 + 2\left(3|n| - 1\right) = 6|n| - 1 = 11 \Rightarrow n = \pm 2$

D) If $f(x) = \left|\left||x| - 2\right| + a\right|$ has exactly three points of non differentiability.

$f(x)$ is non-differentiable at x=0, $|x| - 2 = 0$ or $x = 0, \pm 2$.

Hence, the value of a must be positive, as negative value of a allows $\left||x| - 2\right| + a = 0$ to have real roots, which gives to more points of non differentiability.

50. $(A) \to (p, s); (B) \to (p, r); (C) \to (p, q, r, s)$

(A) $f(0) = RHL = \lim_{x \to 0^+} f(x)$

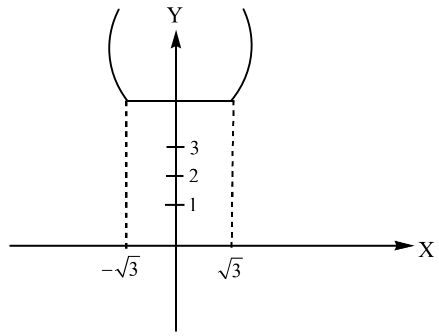
$$= \lim_{h \to 0} f(0 + h) = \lim_{h \to 0} f(h)$$

$$= \lim_{h \to 0}\frac{1 - \cos(1 - \cosh)}{h^4}$$

$$= \lim_{h \to 0}\frac{1 - \cos(1 - \cosh)}{(1 - \cosh)^2}.\lim_{h \to 0}\left(\frac{1 - \cosh}{h^2}\right)^2$$

$$= \frac{1}{2}\cdot\left(\frac{1}{2}\right)^2 = \frac{1}{8} = \frac{1}{\lambda}(given) \quad \because \lambda = 8$$

and $g(x) = \max.\left\{x^2 - 1, 4, x^2 + 1\right\}$



It is clear from graph number of points of non-differentiable is 2. $\because \mu = 2$

then $\lambda + \mu = 10 (p, s)$ is divisible by 2 and 5.

(B) $f(0) = \lim_{n \to \infty}\left(\left(\cos e^n\right)3^{-n^2} + \frac{n^2}{n^2 + n + 1}\right)$

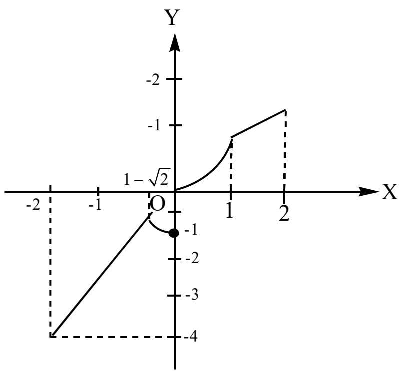
$$= \lim_{n \to \infty}\left(\frac{\left(\cos e^n\right)}{3^{n^n}} + \frac{1}{1 + \frac{1}{n} + \frac{1}{n^2}}\right)$$

$$=\frac{\cos e^{\infty}}{3^{\infty^2}}+\frac{1}{1+0+0}=\frac{-1 to 1}{\infty}+1=0+1 \quad =1$$

$$\therefore \ \lambda=\frac{1}{f(0)}=1$$

and $g(x)=\begin{cases}\min.\{x,x^2\}, & x\ge 0\\ \min.\{2x,x^2-1\}, & x<0\end{cases}$

$$=\begin{cases}2x & ,-2\le x<1-\sqrt{2}\\ x^2-1 & ,1-\sqrt{2}\le x<0\\ x^2 & ,0\le x<1\\ x & ,1\le x\le 2\end{cases}$$



It is clear from the graph g(x) is non-differentiable at 3 points.

$\therefore \ \mu=3$, then $\lambda+\mu=4(p,r)$

is divisible by 2 and 4.

(C) $\because$

$$f\left(\frac{\pi}{4}\right)=RHL=\lim_{x\to\frac{\pi}{4}^+} f(x)=\lim_{h\to 0} f\left(\frac{\pi}{4}+h\right)$$

$$=\lim_{h\to 0}\frac{\sqrt{2}\cos\left(\frac{\pi}{4}+h\right)-1}{\cot\left(\frac{\pi}{4}+h\right)-1}$$

$$=\lim_{h\to 0}\frac{\sqrt{2}\left(\frac{1}{\sqrt{2}}\cos h-\frac{1}{\sqrt{2}}\sin h\right)}{\left(\frac{1-\tan h}{1+\tan h}\right)-1}$$

$$=\lim_{h\to 0}\frac{\left(\frac{-(1-\cos h)}{h}-\frac{\sin h}{h}\right)(1+\tan h)}{-2\left(\frac{\tan h}{h}\right)}$$

$$=\frac{(-0-1)(1+0)}{-2(1)}=\frac{1}{2}=\frac{1}{\lambda}(given)$$

$\therefore \ \lambda=2$ and $g(x)=\dfrac{1-k^{1/x}}{1+k^{1/x}}$ for $0<k<1$

check at x=0

$$LHL==\lim_{x\to 0^-} g(x)=\lim_{h\to 0} g(0-h)$$

$$=\lim_{h\to 0}\frac{1-k^{-1/h}}{1+k^{-1/h}}=\frac{1-0}{1+0}=1$$

$$RHL=\lim_{x\to 0^+} g(x)=\lim_{h\to 0} g(0+h)=\lim_{h\to 0}\frac{1-k^{1/h}}{1+k^{1/h}}$$

$\therefore$ Jump= LHL-RHL=1-(-1)=2

$\therefore \ \mu=2 \ \Rightarrow \ |\lambda-\mu|=0\ (p,q,r,s)$

divisible by 2,3,4 and 5

51. $g(x)$ can be defined as

$$g(x)\begin{cases}f(x),-2\le x<-1\\ -1 \quad ,-1\le x<0\\ f(x),\ 2\le x\le 3\end{cases}$$

or $g(x)\begin{cases}x^2+2x, & -2\le x<-1\\ -1, & -1\le x<0\\ x^2-2x, & 2\le x\le 3\end{cases}$

Clearly $g(x)$ is discontinuous at $x=0$

$g(x)$ is not differentiable at x =0, 2

Ab solute maximum value of $g(x)=f(3)=3$.

52. $(A)\to(p,r)$; $(B)\to(p,q,t)$;

$(C)\to(p,q,r,t)$; $(D)\to(p,s)$

Since $\dfrac{\sin x}{x}\to 1$ as $x\to 0$ but less than 1.

$\therefore \ \dfrac{n_1\sin x}{x}\to n_1$ as $x\to 0$ but less than n_1.

Similary,

$\dfrac{x}{\sin x}\to 1$ as $x\to 0$ but more than or equal to 1.

$\therefore \ \dfrac{n_2 x}{\sin x}\to n_2$ as $x\to 0$ but more than or equal to n_2.

$\therefore\ L=\lim_{x\to0}\left[\frac{n_1\sin x}{x}\right]=n_1-1$

and $M=\lim_{x\to0}\left[\frac{n_2x}{\sin x}\right]=n_2$

Also, $\frac{\tan x}{x}\to1$ as $x\to0$ but more than 1.

$\therefore\ \frac{n_3\tan x}{x}\to n_3$ as $x\to0$ but more than or equal to x_3.

Similarly,

$\frac{x}{\tan x}\to1$ as $x\to0$ but less than 1.

$\therefore\ \frac{n_4x}{\tan x}\to n_4$ as $x\to0$ but less than n_4.

$\Rightarrow N=\lim_{x\to0}\left[\frac{n_3\tan x}{x}\right]=n_3$ and

$P=\lim_{x\to0}\left[\frac{n_4x}{\tan x}\right]=n_4-1$

(A) For $n_1=3,n_4=9$

$\therefore\ \frac{P}{L}=\frac{n_4-1}{n_1-1}=\frac{9-1}{3-1}=4 \qquad (p,r)$

(B) For $n_2=12,n_3=2$

$\therefore\ \frac{M}{N}=\frac{n_2}{n_3}=\frac{12}{2}=6 \quad (p,q,t)$

(C) For $n_2=3,n_4=10$

$\therefore\ P+M=n_4-1+n_2=10-1+3=12$

(D) For $n_1=4,n_2=5,n_3=6,n_4=7$

$\therefore\ L+N+P-M=n_1-1+n_3+n_4-1-n_2$

$=4-1+6+7-1-5=10 \qquad (p,s)$

INTEGER QUESTIONS

53. $f(x)=\begin{cases}e^{-x}, & if\ x\ge0\\ e^{x}, & if\ x>0\end{cases};$

$f'(0^-)=\lim_{h\to0}\frac{f(0-h)-f(0)}{-h}=\lim_{h\to0}\frac{e^{-h}-1}{-h}=1$

ence $\lim_{x\to0}\frac{\left[1-\cos\left(1-\cos\frac{x}{2}\right)\right]}{\left(1-\cos\frac{x}{2}\right)^2}\frac{\left(1-\cos\frac{x}{2}\right)}{2^m.x^n}=1$

Using $\lim_{x\to0}\frac{1-\cos\theta}{\theta^2}=\frac{1}{2}$

$\lim_{x\to0}\frac{\left(1-\cos\frac{x}{2}\right)}{2^m.x^n}=2$, $\lim_{x\to0}\frac{4\sin^4\frac{x}{4}}{2^m.x^n}=2$

$\lim_{x\to0}\frac{4x^4}{4^42^m.x^n}=2 \quad \therefore$ for limit to exist n = 4

$\frac{2}{2^{8+m}}=1,\ 2^{8+m}=2$

$\therefore\ m=-7\Rightarrow n=4$ and $m=-7$

Hence $n^2+m=4^2+(-7)=16-7=9$.

54. i) Continuity should be checked at the end-points of intervals of each definition, i.e., x = 1, 3, 3.5,....

ii) For $\{x^2\}$, continuity should be checked when $x^2=10,11,12$ or $x\sqrt{10},\sqrt{11},\sqrt{12}$,

$\{x^2\}$ is discontinuous for those values of x, where x^2 is an integer.

[Note: Here x^2 is monotonic for the tgiven domain.]

iii) For $\operatorname{sgn}(x-2)$, continuity should be checked when $x-2=0$ or $x=2$

iv) For $[\log_e x]$, continuity should be checked when $\log_e x=1$ or $x=e(\in[1,3])$.

Hence, the overall continuity must be checked at $x=1,2,e,3,\sqrt{10},\sqrt{11},\sqrt{12},3.5$.

Further, $f(1)=0$ and

$\lim_{x\to1^+}f(x)=\lim_{x\to1^+}\operatorname{sgn}(x-2)\times[\log_e x]=0$.

Hence $f(x)$ is continuous at x =1,

$$\lim_{x\to2^-} f(x)=\lim_{x\to2^+}\operatorname{sgn}(x-2)\times[\log_e x]=(-1)\times0=0$$

$$\lim_{x\to2^+} f(x)=\lim_{x\to2^-}\operatorname{sgn}(x-2)\times[\log_e x]=(1)\times0=0$$

Hence, $f(x)$ is continuous at x =2,

$$\lim_{x\to3^-} f(x)=\lim_{x\to3^-}\operatorname{sgn}(x-2)\times[\log_e x]=1$$

$$\Rightarrow \lim_{x\to3^+} f(x)=\lim_{x\to3^+}\{x^2\}=0$$

Hence, $f(x)$ is discontinuous at x =3.

Also $\{x^2\}$ is discontinuous at

$x=\sqrt{10},\sqrt{11},\sqrt{12}$,

Therefore,

$$\lim_{x\to35^-} f(x)=\lim_{x\to3.5^-}\{x^2\}=0.25=f(3.5).$$

Hence, $f(x)$ is discontinuous at

$x=3,\sqrt{10},\sqrt{11},\sqrt{12}$.

55. Putting $x=1+h$, we get

$$f(a)=\lim_{h\to0}\frac{(1+h)^a-a(1+h)+a-1}{h^2}$$

$$=\lim_{h\to0}\frac{\left(1+ah\frac{a(a-1)}{2!}h^2+\ldots\ldots\right)-a-ah+a-1}{h}$$

$$=\frac{a(a-1)}{2}$$

Therefore $f(4)=6$.

56 & 57. Conceptual

58. Let

$$S=\sum_{r=1}^{\infty}\frac{r+2}{2^{r+1}.r.(r+1)}=\sum_{r=1}^{\infty}\frac{2(r+1)-r}{2^{r+1}.r.(r+1)}$$

$$=\sum_{r=1}^{\infty}\frac{1}{2^{r+1}}\left(\frac{2}{r}-\frac{1}{r+1}\right)=\sum_{r=1}^{\infty}\left(\frac{1}{2^r.r}-\frac{1}{2^{r+1}(r+1)}\right)$$

$$=\underset{n\to\infty}{Lim}\left[\left(\frac{1}{2^1.1}-\frac{1}{2^2.2}\right)+\left(\frac{1}{2^2.2}-\frac{1}{2^3.3}\right)\right]+\left(\frac{1}{2^3.3}-\frac{1}{2^4.4}\right)$$

$$+\ldots..+\left(\frac{1}{2^n.n}-\frac{1}{2^{n+1}.(n+1)}\right)$$

$$=\underset{n\to\infty}{Lim}\left(\frac{1}{2}-\frac{1}{2^{n+1}.(n+1)}\right)$$

$\therefore\ S=\frac{1}{2}$ Hence $S^{-1}=2$

59. $$\lim_{x\to1}\left[\frac{-ax+\sin(x-1)+a}{x+\sin(x-1)-1}\right]^{\frac{1-x}{1-\sqrt{x}}}=\frac{1}{4}$$

$$\lim_{x\to1}\left[\frac{\frac{\sin(x-1)}{(x-1)}-a}{\frac{\sin(x-1)}{(x-1)}+1}\right]=\frac{1}{4}\Rightarrow\left(\frac{1-a}{2}\right)^2=\frac{1}{4}$$

$$\Rightarrow\ a=0,a=2\Rightarrow a=2$$

60. It is obvious n is even, then

$$\lim_{n\to\infty}\left(2^{1+3+5+\ldots.+n/2terms}\times3^{2+4+6+\ldots.+n/2terms}\right)^{\frac{1}{(n^2+1)}}$$

$$=\lim_{n\to\infty}\left(2^{\frac{n^2}{4}}\times3^{\frac{n(n+2)}{4}}\right)^{\frac{1}{(n^2+1)}}$$

$$=\lim_{n\to\infty}2^{\frac{n^2}{4(n^2+1)}}\times3^{\frac{n(n+2)}{4(n^2+1)}}$$

$$=2^{\lim\limits_{n\to\infty}\frac{1}{4\left(1+\frac{1}{n^2}\right)}}\times3^{\lim\limits_{n\to\infty}\frac{\left(1+\frac{2}{n}\right)}{4\left(1+\frac{1}{n^2}\right)}}=2^{\frac{1}{4}}3^{\frac{1}{4}}=(6)^{\frac{1}{4}}$$

61. $$\lim_{x\to0}\frac{\sin x^4-x^4\cos x^4+x^{20}}{x^4\left(e^{2x^4}-1-2x^4\right)}$$

$$=\lim_{t\to0}\frac{\sin t-t\cos t+t^5}{t\left(e^{2t}-1-2t\right)}$$

$$= \lim_{t\to 0} \frac{t - \frac{t^3}{3!} + \frac{t^5}{5!} .... - t\left(1 - \frac{t^2}{2!} + \frac{t^4}{4!} ....\right) + t^5}{t\left(1 + 2t + \frac{4t^2}{2!} + \frac{8t^3}{3!} + \frac{16t^4}{4!} + ... - 1 - 2t\right)}$$

$$= \lim_{t\to 0} \frac{-\frac{t^3}{6} + \frac{t^3}{2} + \frac{t^5}{5!} - \frac{t^5}{4!} + ... + t^5}{2t^3 + \frac{8t^4}{3!} + ...}$$

$$= \frac{-\frac{1}{6} + \frac{1}{2}}{2} = -\frac{-1+3}{12} = \frac{1}{6}$$

62. $\lim_{x\to 0} \sin\left(\frac{\pi\left(1-\cos^m x\right)}{x^n}\right)$

$$= \sin\left(\lim_{x\to 0} \frac{\pi\left(1-\cos^m x\right)}{x^n}\right)$$

$$= \sin\left(\lim_{x\to 0} 2\pi m \frac{\sin^2 x/2}{x^n}\right)$$

$\Rightarrow m \in N$ and $n = 1$ or 2

63. $L = e^{\lim_{x\to\infty} \frac{\ln\left(e^{2x}+e^x+x\right)}{x}}$

using L Hospital's rule

$L = e^{\lim_{x\to\infty} \frac{2e^{2x}+e^x+1}{e^{2x}+e^x+x}} = e^{\lim_{x\to\infty} \frac{2+e^{-x}+e^{-2x}}{1+e^{-x}+xe^{-2x}}} = e^2$

64. $\sin^{-1} x$ is a monotonically increasing function. Hence, $f(x) = \left[\sin^{-1} x\right]$ is discontinuous, where $\sin^{-1} x$ is an integer.

$\Rightarrow \sin^{-1} x = -1, 0, 1$ or $x = -\sin 1, 0, \sin 1$

$\frac{2}{1+x^2}, x \geq 0$, is a monotonically decreasing function.

Hence, $f(x) = \left[\frac{2}{1+x^2}\right], x \geq 0$ is discontinuous when $\frac{2}{1+x^2}$ is an integer.

$\Rightarrow \frac{2}{1+x^2} = 1, 2 \Rightarrow x - 1, 0$

65. $\sin(x-2)$ and $\cos(x-2)$ are continuous for all x. Since $\left[x^3\right]$ is not continuous at integral values of x^3, $f(x)$ is continous in $[0,2]$ if $\left[\frac{(x-2)^3}{a}\right] = 0$, $\forall x \in [0,2]$.

Now, $(x-2)^3 \in [0,8]$ for $x \in [4,6]$

$\Rightarrow a > 8$ for $\left[\frac{(x-2)^3}{a}\right] = 0$

66. $f(x) = \begin{cases} x-1, & x \text{ is rational} \\ x^2 - x - 2, & x \text{ is irrational} \end{cases}$ is continuous when $x - 1 = x^2 - x - 2$

or $x^2 - 2x - 1 = 0$ or $x = \frac{2 \pm \sqrt{8}}{2}$.

Hence, $f(x)$ is continuous at two points $\Rightarrow a = 2$.

$f(x) = \text{sgn}\left(x^3 - 3x + 1\right)$ is discontinuous when $x^3 - 3x + 1 = 0$.

Now, $y = x^3 - 3x + 1$ has three distinct roots as $\frac{dy}{dx} = 3x^2 - 3 = 0 \Rightarrow x \pm 1$

Also $f(1) = 1 - 3 + 1 = -1$ and $f(-1) = -1 + 3 + 1 = 3$. Hence, the graph of $y = x^3 - 3x + 1$ is

Hence, $f(x) = \text{sgn}\left(x^3 - 3x + 1\right)$ is discontinuous at three points $\Rightarrow b = 3$.

$f(x) = (\log x)\left|x^2 - 4x + 3\right| + 2(x-2)^{1/3}$

$= (\log x)|x-1||x-3| + 2(x-2)^{1/3}$

Which is non-differentiable at $x = 3$ and $x = 2$ as at $x = 3$, $f(x)$ has a sharp turn and at $x = 2$, $f(x)$ have a vertical tangent.

At $x = 1$, $f(x)$ is differentiable. So $c = 2$ and $d = 1$

67. $f(x)=\lim\limits_{n\to\infty}\dfrac{(x^2)^n-1}{(x^2)^n+1}$

$$=\lim_{n\to\infty}\frac{1-\dfrac{1}{(x^2)^n}}{1+\dfrac{1}{(x^2)^n}}$$

$$=\begin{cases}-1, & 0\le x^2<1\\ 0, & x^2=1\\ 1, & x^2>1\end{cases}=\begin{cases}1, & x<-1\\ 0, & x=-1\\ -1, & -1<x<1\\ 0, & x=1\\ 1, & x>1\end{cases}$$

Thus $f(x)$ is discontinuous at $x=\pm1$ .69.

$f(x)=\lim\limits_{n\to\infty}\dfrac{(x^2)^n-1}{(x^2)^n+1}$

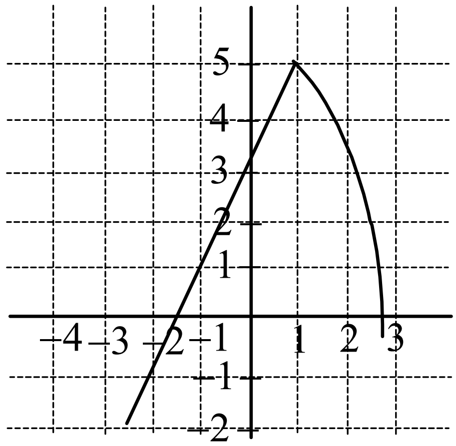
$$=\lim_{n\to\infty}\frac{1-\dfrac{1}{(x^2)^n}}{1+\dfrac{1}{(x^2)^n}}$$

$$=\begin{cases}-1, & 0\le x^2<1\\ 0, & x^2=1\\ 1, & x^2>1\end{cases}=\begin{cases}1, & x<-1\\ 0, & x=-1\\ -1, & -1<x<1\\ 0, & x=1\\ 1, & x>1\end{cases}$$

Thus $f(x)$ is discontinuous at $x=\pm1$.

68.

Graph of $y=f(x)$



Graph of $y=f(|x|)$